

Introduction to the ab initio nuclear many-body problem

Lecture 4: *Quantum Monte Carlo as an ab initio method*

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May 27, 2026

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1 The Quantum Monte Carlo Family of approaches

2 Preliminaries from numerical analysis: the Monte Carlo quadrature

Many concepts discussed in this Lecture can be understood purely from a numerical analysis perspective: we are integrating Schrödinger's equation. To facilitate the discussion we will start from the essential elements of the Monte Carlo quadrature.

Monte Carlo relies on estimating integrals by interpreting them as expectations of statistical distributions; so let's start from there. We'll start from the case of a single variable and generalize to many variables, where Monte Carlo integration has its edge. The expectation of a function f of a continuous random variable X distributed according to a probability distribution $p(x)$.

$$\langle f(X) \rangle = \int_{-\infty}^{\infty} dx p(x) f(x) \quad (1)$$

we will take $p(x) = w(x)/(b-a)$ for $x \in [a, b]$ and zero elsewhere; this is sufficient for most practical purposes since even when the integration extends to infinity, often in practice one integrates only up to some large number. This brings the expectation to the form

$$\langle f(X) \rangle = \frac{1}{b-a} \int_a^b dx w(x) f(x) \quad (2)$$

This is of the form of a general Gaussian quadrature where the integral is subsequently discretized on a set of points that correspond to extrema or nodes of a family of orthogonal polynomials. Here we'll take an alternative approach and by interpreting $w(x)$ as a probability density we will evaluate the integral stochastically. Taking the definition of a variance of a random variable we can also define

$$\text{var}[f(X)] = \langle [f(x) - \langle f(X) \rangle]^2 \rangle = \langle f(X)^2 \rangle - \langle f(X) \rangle^2 \quad (3)$$

$$= \frac{1}{b-a} \int_a^b dx w(x) f^2(x) - \left(\frac{1}{b-a} \int_a^b dx w(x) f(x) \right)^2 \quad (4)$$

These two quantities, $\langle f(X) \rangle$ and $\text{var}[f(X)]$, are named *population mean* and *population variance*, respectively, and they are the ones we wish to calculate. The former will be the value of the integral we are after and the latter will define the error of our estimation via the population standard deviation $\sqrt{\text{var}[f(X)]}$. To estimate these quantities we will define what is usually referred to as *estimators*, that is, recipes for estimating a statistical property of a population using observed data; in our case the observed data will be evaluations of the function $f(x)$ for a sample of values for x .

Assume a sample of n random variables $X_1, X_2, \dots, X_n$ are drawn from $w(x)$. Applying the function f to each one of them leads to a new sample of random variables $f(X_1), f(X_2), \dots, f(X_n)$. We call the *sample mean* the mean of the second sample

$$\bar{f} = \frac{1}{n} \sum_{i=1}^n f(X_i) . \quad (5)$$

Making a distinction as to what is considered a random variable here is important. The variables X_i are random variables and so are $f(X_i)$ since the function of a random variable is also a random variable (albeit following a different distribution). Consequently $\bar{f}$ is also a random variable as it is the sum of the random variables $f(X_i)$, the

sum being, of course, a function. Since $\bar{f}$ is a random variable we can calculate its expectation

$$\langle \bar{f} \rangle = \left\langle \frac{1}{n} \sum_{i=1}^n f(X_i) \right\rangle = \frac{1}{n} \sum_{i=1}^n \langle f(X_i) \rangle = \langle f(X) \rangle . \quad (6)$$

Note that since the sample mean is computed by a finite number of samples n , $\bar{f}$ itself could be quite far from $\langle f(X) \rangle$ (e.g., consider the case $n = 2$). However, Eq. (6) means that the *expectation* of the sample mean is equal to the population mean. Even better, one can show that as n grows larger the probability of $\bar{f}$ being equal to $\langle f(X) \rangle$ increases. This motivates us to choose $\bar{f}$ as an estimator for $\langle f(X) \rangle$, and consequently as an approximation to our integral. How good of an approximation $\bar{f}$ is for $\langle f(X) \rangle$ or our integral is related to the variance of the sample mean (again, remember, the sample mean is a random variable):

$$\text{var}[\bar{f}] = \text{var} \left[\frac{1}{n} \sum_{i=1}^n f(X_i) \right] = \frac{1}{n^2} \sum_{i=1}^n \text{var}[f(X)] = \frac{1}{n} \text{var}[f(X)] , \quad (7)$$

where we have used the identity for the variance of a sum of random variables. Hence the standard deviation of the sample mean, tells of the error made the sample mean as an estimator, is

$$\sqrt{\text{var}[\bar{f}]} = \frac{1}{\sqrt{n}} \sqrt{\text{var}[f(X)]} . \quad (8)$$

This result is at the heart of Monte Carlo quadrature's success in evaluating multidimensional integrals. The variance drops as $1/\sqrt{n}$ and that scaling doesn't depend on the space's dimensionality. Hence the same efficiency is attained in higher dimensional integrals as well, beating all other methods; more on this later.

We'll now play the same game with $\text{var}[f(X)]$ and find an estimator for it. Intuitively, we turn to the sample variance

$$e_{\text{var}} = \bar{f}^2 - \bar{f}^2 = \frac{1}{n} \sum_{i=1}^n f^2(X_i) - \left[\frac{1}{n} \sum_{i=1}^n f(X_i) \right]^2 . \quad (9)$$

Again, being explicit: e_{var} is a function of random variable and hence a random variable itself. its expectation is

$$\langle e_{\text{var}} \rangle = \frac{n-1}{n} \text{var}[f(X)] . \quad (10)$$

So we can use the sample variance (e_{var}) to define an estimator for the population variance ($\text{var}[f(X)]$) which in turn will allow us to evaluate the error made by the sample mean ($\bar{f}$) as an estimator for population mean ($\langle f(X) \rangle$), which is our integral. Putting everything together the Monte Carlo estimation for our integral is

$$\int_a^b dx w(x) f(x) \approx (b-a) \bar{f} \pm (b-a) \sqrt{\text{var}[\bar{f}]} \quad (11)$$

$$\approx \frac{(b-a)}{n} \sum_{i=1}^n f(X_i) \pm \frac{b-a}{\sqrt{n-1}} \sqrt{\frac{1}{n} \sum_{i=1}^n f^2(X_i) - \left[\frac{1}{n} \sum_{i=1}^n f(X_i) \right]^2} \quad (12)$$

where we employed the central limit theorem that allows us to treat $\bar{f}$ as a Gaussian distributed variable for large n , regardless of what $w(x)$ was used to draw the samples of X_i . The generalization to d -dimensions is straightforward and results in promoting $(b-a)$ to a d -dimensional volume V leading to the prescription

$$\int_V d^d x w(x) f(\mathbf{x}) \approx V \bar{f} \pm V \sqrt{\text{var}[\bar{f}]} \quad (13)$$

$$\approx \frac{V}{n} \sum_{i=1}^n f(X_i) \pm \frac{V}{\sqrt{n-1}} \sqrt{\frac{1}{n} \sum_{i=1}^n f^2(X_i) - \left[\frac{1}{n} \sum_{i=1}^n f(X_i) \right]^2} \quad (14)$$

3 Variational Monte Carlo (VMC)

3.1 Introduction

Variational Monte Carlo (VMC) is typically formulated in coordinate space. We will first set up the appropriate language to describe systems in coordinate space, and use it to express the Hamiltonian. Then we will discuss the concept of a trial wavefunction, that is, our best guess for the Hamiltonian's ground state. Finally we will reformulate Schrödinger's equation in a variational form.

3.2 The QMC-*esque* coordinate space language

A system of N particles in d dimensions is described using Nd -dimensional vector of all positions of all particles

$$\mathbf{X} = (r_{11}, r_{12}, \dots, r_{1d}, \dots, r_{N1}, r_{N2}, \dots, r_{Nd})^T, \quad (15)$$

where r_{ij} is the j -th coordinate of the i -th particle, e.g., the second (i.e., y coordinate) of the 6th particle is r_{62} . For example, for 2 particles in 3 dimensions ($N = 2, d = 3$),

$$\mathbf{X} = (r_{11}, r_{12}, r_{13}, r_{21}, r_{22}, r_{23})^T. \quad (16)$$

We are after the *ground state energy* of the system, E_0 . This is usually the point of departure for any other observable. Variational Monte Carlo provides an upper limit to E_0 .

3.3 The Hamiltonian

The many-body Hamiltonian can be written in coordinate space

$$\hat{H} = \sum_{i=1}^N \hat{K}_i + \sum_{i<j} \hat{V}_{ij}^{(2)} + \sum_{i<j<k=1}^N \hat{V}_{ijk}^{(3)} + \dots \quad (17)$$

$$\hat{H} = \sum_{i=1}^N \left(-\frac{\hbar^2}{2m} \nabla_i^2 \right) + \sum_{i<j=1}^N V^{(2)}(\mathbf{r}_i, \mathbf{r}_j) + \sum_{i<j<k=1}^N V^{(3)}(\mathbf{r}_i, \mathbf{r}_j, \mathbf{r}_k) + \dots \quad (18)$$

$$\hat{H} = \sum_{i=1}^N \left(-\frac{\hbar^2}{2m} \nabla_i^2 \right) + \sum_{i<j=1}^N V^{(2)}(\mathbf{r}_i, \mathbf{r}_j) + \sum_{i<j<k=1}^N V^{(3)}(\mathbf{r}_i, \mathbf{r}_j, \mathbf{r}_k) + \dots \quad (19)$$

$$(20)$$

where ∇_i is the vector $(\partial_{r_{i1}}, \partial_{r_{i2}}, \dots, \partial_{r_{id}})$ acting on the coordinates of particle i . The superscript of the operators $V^{(n)}$ signifies an n -body interaction: in QMC methods dealing with $N > 2$ -body interactions is relatively straightforward, which is a relief for the study of nuclear systems (see Lecture 3).

Why is there $i < j$ in the sum of two-body interactions?

The form of the Hamiltonian, e.g., in Eq. (19), guides the choice for a trial wavefunction. Inspired by ideas from perturbation theory, we could write the Hamiltonian as $\hat{H} = \hat{H}_0 + \hat{H}_1$, where $\hat{H}_0$'s ground state can be easily determined, and use that as a trial wave-function. Alternatively, we could use a phenomenological approximation to the ground state of $\hat{H}$ for a trial wavefunction. More details, below.

Finally, to pose the problem well, we need to specify the boundary conditions that are applied to the particles. Note that even though we will not be solving and differential equations directly, the boundary conditions will enter our calculations.

Identify where the boundary conditions enter in the VMC formulation

3.4 Trial wavefunction

The trial wavefunction is our best guess for how the ground-state of Eq. (19) looks like. *One cannot prove within VMC that the guess is exactly correct.* However, due to the variational principle, we can guarantee that the energy calculated with it is an upper bound of the ground-state energy. This creates a practical strategy: one can keep updating the guesses until the calculated energy doesn't change. As we will see later, this can be automatized to an extent. In what follows we'll go over some standard choices for trial-wavefunctions.

A slater determinant As already mentioned above a standard choice for the ground-state's form is the Slater Determinant (SD); see Lecture 1. This corresponds to perturbation-theory inspired strategy mentioned above, i.e., splitting into two parts $\hat{H} = \hat{H}_0 + \hat{H}_1$, with $\hat{H}_0 = \hat{K}$ and using eigenstates of $\hat{H}_0$ as trial states:

$$\Phi_0^{(\text{SD})}(\mathbf{X}; \mathbf{k}_1, \mathbf{k}_2, \dots, \mathbf{k}_N) = \mathcal{A}[\phi_{\mathbf{k}_1}(\mathbf{r}_1), \phi_{\mathbf{k}_2}(\mathbf{r}_2), \dots, \phi_{\mathbf{k}_N}(\mathbf{r}_N)] = \det \begin{pmatrix} \phi_{\mathbf{k}_1}(\mathbf{r}_1) & \phi_{\mathbf{k}_1}(\mathbf{r}_2) & \cdots & \phi_{\mathbf{k}_1}(\mathbf{r}_N) \\ \phi_{\mathbf{k}_2}(\mathbf{r}_1) & \phi_{\mathbf{k}_2}(\mathbf{r}_2) & \cdots & \phi_{\mathbf{k}_2}(\mathbf{r}_N) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_{\mathbf{k}_N}(\mathbf{r}_1) & \phi_{\mathbf{k}_N}(\mathbf{r}_2) & \cdots & \phi_{\mathbf{k}_N}(\mathbf{r}_N) \end{pmatrix} \quad (21)$$

where $\phi_{\mathbf{k}_i}(\mathbf{r}_j)$ is an eigenstate of K with energy $E_i = \hbar^2 k_i^2 / 2m$ evaluated at the coordinate of particle j :

$$K\phi_{\mathbf{k}_i}(\mathbf{r}) = \frac{\hbar^2}{2m} k_i^2 \phi_{\mathbf{k}_i}(\mathbf{r}) \quad (22)$$

In the case of a spin-balanced system of N particles, that is, with $N/2$ particles per species (spin-up and spin-down)¹, we can generalize this trial wavefunction to the product of two SDs:

$$\Psi_0^{(\text{SD})}(\mathbf{X}; \mathbf{k}_1, \mathbf{k}_2, \dots, \mathbf{k}_{N/2}, \mathbf{k}_{1'}, \mathbf{k}_{2'}, \dots, \mathbf{k}_{N/2'}) = \Phi_0^{(\text{SD})}(\mathbf{k}_1, \mathbf{k}_2, \dots, \mathbf{k}_{N/2}) \Phi_0^{(\text{SD})}(\mathbf{k}_{1'}, \mathbf{k}_{2'}, \dots, \mathbf{k}_{N/2'}) \quad (23)$$

$$= \mathcal{A}[\phi_{\mathbf{k}_{1'}}(\mathbf{r}_{1'}), \phi_{\mathbf{k}_{2'}}(\mathbf{r}_{2'}), \dots, \phi_{\mathbf{k}_{N/2'}}(\mathbf{r}_{N/2'})] \mathcal{A}[\phi_{\mathbf{k}_1}(\mathbf{r}_1), \phi_{\mathbf{k}_2}(\mathbf{r}_2), \dots, \phi_{\mathbf{k}_{N/2}}(\mathbf{r}_{N/2})] \quad (24)$$

$$= \det \begin{pmatrix} \phi_{\mathbf{k}_1}(\mathbf{r}_1) & \phi_{\mathbf{k}_1}(\mathbf{r}_2) & \cdots & \phi_{\mathbf{k}_1}(\mathbf{r}_N) \\ \phi_{\mathbf{k}_2}(\mathbf{r}_1) & \phi_{\mathbf{k}_2}(\mathbf{r}_2) & \cdots & \phi_{\mathbf{k}_2}(\mathbf{r}_N) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_{\mathbf{k}_N}(\mathbf{r}_1) & \phi_{\mathbf{k}_N}(\mathbf{r}_2) & \cdots & \phi_{\mathbf{k}_N}(\mathbf{r}_N) \end{pmatrix} \det \begin{pmatrix} \phi_{\mathbf{k}_{1'}}(\mathbf{r}_{1'}) & \phi_{\mathbf{k}_{1'}}(\mathbf{r}_{2'}) & \cdots & \phi_{\mathbf{k}_{1'}}(\mathbf{r}_{N'}) \\ \phi_{\mathbf{k}_{2'}}(\mathbf{r}_{1'}) & \phi_{\mathbf{k}_{2'}}(\mathbf{r}_{2'}) & \cdots & \phi_{\mathbf{k}_{2'}}(\mathbf{r}_{N'}) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_{\mathbf{k}_{N'}}(\mathbf{r}_{1'}) & \phi_{\mathbf{k}_{N'}}(\mathbf{r}_{2'}) & \cdots & \phi_{\mathbf{k}_{N'}}(\mathbf{r}_{N'}) \end{pmatrix} \quad (25)$$

A BCS determinant A state with the lowest range of many-particle correlations is the BCS state which can be written in the determinant form (see Lecture 1&2)

$$\Psi_0^{(\text{BCS})}(\mathbf{X}; \varphi) = \mathcal{A}[\varphi(\mathbf{r}_1, \mathbf{r}_2), \varphi(\mathbf{r}_3, \mathbf{r}_4), \dots, \varphi(\mathbf{r}_{N-1}, \mathbf{r}_N)] = \det \begin{pmatrix} \varphi(\mathbf{r}_1, \mathbf{r}_{1'}) & \varphi(\mathbf{r}_1, \mathbf{r}_{2'}) & \cdots & \varphi(\mathbf{r}_1, \mathbf{r}_{N'/2}) \\ \varphi(\mathbf{r}_2, \mathbf{r}_{1'}) & \varphi(\mathbf{r}_2, \mathbf{r}_{2'}) & \cdots & \varphi(\mathbf{r}_2, \mathbf{r}_{N'/2}) \\ \vdots & \ddots & \ddots & \vdots \\ \varphi(\mathbf{r}_{N/2}, \mathbf{r}_{1'}) & \varphi(\mathbf{r}_{N/2}, \mathbf{r}_{2'}) & \cdots & \varphi(\mathbf{r}_{N/2}, \mathbf{r}_{N'/2}) \end{pmatrix} \quad (26)$$

which depends functionall on the *pair wavefunctions*,

$$\varphi(\mathbf{r}_i, \mathbf{r}_{j'}; \{c_n\}, t) = \sum_n c_n \phi_{\mathbf{k}_n}(\mathbf{r}_i) \phi_{t(\mathbf{k}_n)}(\mathbf{r}_{j'}) \quad (27)$$

¹In the absence of spin-spin interactions, no spin-flips can occur and it is common to refer to each spin population as “species”.

which is in turned defined by the pairing amplitudes c_n and the *pairing scheme* t which pairs single-particle state mfk to state $t(\mathbf{k})$. Note that instead of pairing momentum states one could also pair linear combination of them in the generalized pair wavefunction

$$\tilde{\varphi}(\mathbf{r}_i, \mathbf{r}_{j'}; \{c_n\}, t) = \sum_n \tilde{c}_n g_n(\mathbf{r}_i) g_{t(n)}(\mathbf{r}_{j'}) \quad (28)$$

where

$$g_n(\mathbf{r}) = \sum_m a_{nm} \phi_{\mathbf{k}_m}(\mathbf{r}) \quad (29)$$

This is more similar to the HFB state discussed in Lecture 2.

For example, in the simplest case of the singlet-pairing encountered in terrestrial superconductors, the BCS pair state in Eq. (27) is a good phenomenological model with the pairing scheme t corresponding to the time-reversal operator and the pairing amplitudes from BCS theory,

$$t(\mathbf{k}) = -\mathbf{k}, \quad c_n = \frac{v(\mathbf{k})}{u(\mathbf{k})} \quad (30)$$

3.5 Recasting Schrodinger's Equation: the Rayleigh-Ritz principle

3.5.1 The general formulation

At the heart of VMC lies the fact that the expectation value of the Hamiltonian at *any* trial state ψ_T is an upper bound to the ground state. This is a general result that holds for any ψ_T , but, of course, its practical value is seen when ψ_T is chosen “close” to the ground state and its energy becomes a tight bound on the ground state energy. What is meant by “close” here will be quantified in this section to mean “large overlap in Hilbert space”. Furthermore, we will validate the assumption that a the “closer” a trial state is to the ground state, the tighter the energy bound.

First we define the basis spanned by eigenstates of the many-body Hamiltonian in Eq. (19), $|\Phi_n\rangle$, that is,

$$H |\Phi_n\rangle = E_n |\Phi_n\rangle .$$

Even though we don't know much about $|\Phi_n\rangle$ (after all our goal here is to acquire access to $|\Phi_0\rangle$, or at least E_0), we will use that their eigenstates of H , their orthonormality, $\langle \Phi_n | \Phi_m \rangle = \delta_{nm}$ and their completeness, i.e., that they form a basis, for a proof. Our trial state can be expressed as

$$|\Psi_T\rangle = \sum_{n=0}^{\infty} a_n |\Phi_n\rangle, \quad \text{with } a_n = \langle \Phi_n | \Psi_T \rangle \quad (31)$$

Note that we don't assume that $|\psi_T\rangle$ is normalized. The so-called variational energy E_V is just the normalized expectation value of the Hamiltonian at $|\Psi_T\rangle$,

$$E_V = \frac{\langle \Psi_T | H | \Psi_T \rangle}{\langle \Psi_T | \Psi_T \rangle} = \frac{\sum_{n,n'} a_n^* a_n \langle \Phi_{n'} | H | \Phi_n \rangle}{\sum_{n,n'} a_n^* a_n \langle \Phi_{n'} | \Phi_n \rangle} = \frac{\sum_n |a_n|^2 E_n}{\sum_n |a_n|^2} = E_0 + \frac{\sum_n |a_n|^2 (E_n - E_0)}{\sum_n |a_n|^2} \geq E_0 \quad (32)$$

where to prove the bound we have used that $E_0 < E_n$ for every n . This result also allows us to use the variational energy to quantify when a state is “close” to the ground state. To see this more concretely, let's consider our trial state to be normalized $\langle \Psi_T | \Psi_T \rangle = 1$, or $\sum_n |a_n|^2 = 1$ (equivalently we could define $\tilde{a}_n = a_n / \sqrt{\sum_n |a_n|^2}$), such that the bound we just proved becomes

$$E_V = E_0 + \sum_n |a_n|^2 (E_n - E_0) \quad (33)$$

We will show that a small variation away from the ground state translates to a small shift in energy. Indeed, if our trial state is almost equal to the ground state, with only a small admixture of a state $|\Phi_1\rangle$, then $a_1 = \eta$, $a_0 = \sqrt{1 - \eta^2}$, and $a_n = 0$ otherwise, where we are implying that η is a small number. This means,

$$|\Psi_T\rangle = \sum_{n=0}^{\infty} a_n |\Phi_n\rangle = \sqrt{1 - \eta^2} |\Phi_0\rangle + \eta |\Phi_1\rangle \quad (34)$$

and the bound becomes

$$E_V = E_0 + \sum_n |a_n|^2 (E_n - E_0) = E_0 + \eta^2 (E_1 - E_0) \quad (35)$$

In words, as the excited-state components of the trial state decrease *linearly* the variational energy approaches the ground state energy *quadratically*. So, the variational energy can quantify the distance of the trial state from the ground state. This provides the motivation behind VMC: stepping through a Hilbert space in a direction that decreases the variational energy is a path that leads to the ground state. If one parametrizes such a walk with a set of parameters α_i , such that $|\Psi_T(\alpha_i)\rangle$ is a state along this walk, finding the best approximation to the ground state is reduced to a standard *multi-dimensional* minimization problem that involves minimizing the function

$$E_V(\alpha) = \frac{\langle \Psi_T(\alpha) | H | \Psi_T(\alpha) \rangle}{\langle \Psi_T(\alpha) | \Psi_T(\alpha) \rangle} \quad (36)$$

with respect to the parameters α .

Here we have hidden the complexity of providing good approximations to the many-body ground state in the work needed for a single evaluation of the function $E_V(\alpha)$. Indeed, the RHS of Eq. (36) contains two integrals that are in general very hard to be evaluated numerically:

$$E_V(\alpha) = \frac{\langle \Psi_T(\alpha) | H | \Psi_T(\alpha) \rangle}{\langle \Psi_T(\alpha) | \Psi_T(\alpha) \rangle} = \frac{\int d^d r \Psi_T^*(\mathbf{r}; \alpha) H \Psi_T(\mathbf{r}; \alpha)}{\int d^d r |\Psi_T(\mathbf{r}; \alpha)|^2} \quad (37)$$

The success of VMC lies in its efficiency to evaluate these integrals using Monte Carlo techniques.

3.5.2 The variational principle for excited states

An important caveat to the above proof is that E_V is an *upper bound to the lowest eigenvalue whose eigenstate is not orthogonal to $|\Psi_T\rangle$* . This should be straightforward to see from Eq. (32) where if $|\Psi_T\rangle$ were orthogonal to $|\Phi_0\rangle$ then E_0 would be absent because it would be $a_0 = 0$. In that case we would get a bound for E_1 (similarly, if $\langle \Psi_T | \Phi_1 \rangle \neq 0$. In the case where $|\Psi_T\rangle$ depends on variational parameters α , then the bound holds if for there is α_0 such that $\langle \Psi_T(\alpha_0) | \Phi_0 \rangle \neq 0$. Of course this hold in principle, in practice it's better if there is a finite (and preferably large) region in the α -space in which $|\Psi_T\rangle$ is not orthogonal to $|\Phi_0\rangle$. This is because in practice one needs to survey (at least some part of) the space of possible α 's to identify the minimum: isolate minima are hard to identify. This is where physical intuition is needed to identify the expected symmetries of the ground state and ensure that the trial state chosen has components that posses those symmetries.

While this caveat can be a nuisance when looking for the ground state, one can take advantage of it to use the variational principle to provide an upper bound for excited states of the Hamiltonian. This is done when by some physical principle we are certain of some symmetry of the ground state. Then we can construct a trial state $|\Psi_T\rangle$ that has no components with this symmetry guaranteeing that $\langle \Psi_T | \Phi_0 \rangle = 0$

Note that for this to work then it should be $\langle \Psi(\alpha) | \Phi_0 \rangle = 0$ for all possible values of α . This might sound trivial but we'll see in the example below how it can become a subtle point.

We will see this play out clearly in the example below.

3.5.3 Intuitive example from matrix diagonalization

In the above proof we have employed no physical argument and so the variational principle should hold for any matrix diagonalization: given a matrix O and a vector $\mathbf{v}$, the number $\mathbf{v}^\dagger O \mathbf{v} / \mathbf{v}^\dagger \cdot \mathbf{v}$ is greater or equal to the lowest eigenvalue of O that is not orthogonal to $\mathbf{v}$. We'll see this for the special case of a symmetric 2×2 matrix.

Consider the symmetric matrix

$$O = \begin{pmatrix} O_{11} & O_{12} \\ O_{21} & O_{22} \end{pmatrix}, \quad (38)$$

with $O_{12} = O_{21}$. The diagonalization of this matrix is trivial and so we will use it as an example to get an insight on how the variational principle works. First we will diagonalize O by identifying the unitary transformation that makes it diagonal and then apply the variational principle to get an upper bound to the lowest eigenvalue. By doing so we'll see that if by some other considerations one knows enough about the symmetries of the lowest eigenvalue's vector, the variational principle can be invoked to identify it *exactly*. Let U be the unitary matrix that diagonalizes O , that is,

$$U^\dagger O U = \begin{pmatrix} \omega_1 & 0 \\ 0 & \omega_2 \end{pmatrix}, \quad (39)$$

where ω_i are the eigenvalues of O . The requirement for U to be unitary provides three constraints for the elements of U :

$$U^\dagger U = \begin{pmatrix} U_{11}U_{11} + U_{21}U_{21} & U_{11}U_{12} + U_{21}U_{22} \\ U_{11}U_{12} + U_{21}U_{22} & U_{12}U_{12} + U_{22}U_{22} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \implies \begin{aligned} U_{11}^2 + U_{21}^2 &= 1 \\ U_{11}U_{12} + U_{21}U_{22} &= 0 \\ U_{12}^2 + U_{22}^2 &= 1 \end{aligned} \quad (40)$$

This means that U is described by a single parameter. It's then convenient to write U in terms of a parameter θ as

$$U(\theta) = \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix} \quad (41)$$

which satisfies all constraints and it's the most general form of an orthogonal matrix. Then it is

$$U^\dagger O U = \begin{pmatrix} O_{11} \cos^2 \theta + O_{22} \sin^2 \theta + O_{12} \sin 2\theta & \frac{1}{2}(O_{11} - O_{22}) \sin 2\theta - O_{12} \cos 2\theta \\ \frac{1}{2}(O_{11} - O_{22}) \sin 2\theta - O_{12} \cos 2\theta & O_{11} \cos^2 \theta + O_{22} \sin^2 \theta - O_{12} \sin 2\theta \end{pmatrix} \quad (42)$$

We can now choose $\theta = \theta_0$ such that this matrix becomes diagonal, that is, its off-diagonal elements, which are both the same, vanish, or

$$\frac{1}{2}(O_{11} - O_{22}) \sin 2\theta_0 - O_{12} \cos 2\theta_0 = 0. \quad (43)$$

Then we can read the eigenvalues from the diagonal elements,

$$\omega_{\frac{1}{2}} = O_{11} \cos^2 \theta + O_{22} \sin^2 \theta \pm O_{12} \sin 2\theta \quad (44)$$

Note that this reproduces the known behaviour of the bifurcation of eigenvalues of a 2×2 symmetric matrix. The solution of Eq. (43),

$$\theta_0 = \frac{1}{2} \arctan \left(\frac{2O_{12}}{O_{11} - O_{22}} \right) \quad (45)$$

Further, we know that the columns of the transformation matrix U correspond to the eigenvectors and so

$$\mathbf{v}_1 = (\cos \theta_0, \sin \theta_0)^T, \quad \mathbf{v}_2 = (\sin \theta_0, -\cos \theta_0)^T, \quad (46)$$

and the diagonalization of O is done. Note that $\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$, as expected for the eigenvectors of a Hermitian matrix.

We will now pretend that we don't know the eigenvalues of O and apply the variational principle to get an upper bound to the lowest one. We choose for our $\mathbf{v}_T$ the normalized vector

$$\mathbf{v}(\phi) = (\cos \phi, \sin \phi)^T. \quad (47)$$

Then we calculate

$$\omega(\phi) = \frac{\mathbf{v}(\phi)^\dagger O \mathbf{v}(\phi)}{\mathbf{v}(\phi)^\dagger \cdot \mathbf{v}(\phi)} = \sin^2(\phi) O_{11} - 2 \cos(\phi) \sin(\phi) O_{12} + \cos^2(\phi) O_{22} \quad (48)$$

$$= \sin^2(\phi) O_{11} + \sin(2\phi) O_{12} + \cos^2(\phi) O_{22} \quad (49)$$

We can now minimize this function of a single variable by finding the roots of its derivative

$$\omega'(\phi) = 2 \cos(2\phi) O_{12} - 2 \cos(\phi) \sin(\phi) (O_{11} - O_{22}) \quad (50)$$

and solving for the root ϕ_0 gives

$$\omega'(\phi_0) = 0 \implies \phi_0 = \frac{1}{2} \arctan \left(\frac{2O_{12}}{O_{11} - O_{22}} \right) = \theta_0 \quad (51)$$

That is we found the special case where the equal sign holds in Eq. (32). Now $\omega(\phi_0)$ is the lowest eigenvalue and $\mathbf{v}(\phi_0)$ is the best approximation of the corresponding eigenstate, which in this case is the eigenstate itself.

Given the above discussion now one might be tempted to use $\tilde{\mathbf{v}}(\phi) = (\sin \phi, -\cos \phi)$ to get the other eigenvalue since

$$\tilde{\mathbf{v}}^\dagger(\phi) \cdot \mathbf{v}(\phi) = 0 \quad (52)$$

for every ϕ . However, this would lead to the same ϕ_0 . This is because

$$\tilde{\mathbf{v}}^\dagger(\phi) \cdot \mathbf{v}_1(\theta_0) = \sin(\phi) \cos(\theta_0) - \cos(\phi) \sin(\theta_0) \quad (53)$$

which is 0 only for $\phi = \theta_0$.

Try multiplying with $1/(\phi - \theta_0)$

3.5.4 Why Monte Carlo?

There is nothing in the Variational form of Schrödinger's equation that prohibits the use of your favourite technique for the evaluation of the integrals. One could try to evaluate Eq. (?) using e.g., Simpson's rule. When interested in one-particle in 1, 2, or 3 dimensions, this might suffice. However, in the case of many-body systems, this would lead one to discover the curse of dimensionality: when using a quadrature rule whose power of the leading error is p , the integration error for a d dimensional integral scales with the number of function evaluations $\mathcal{N}$ we're willing to do as $\mathcal{N}^{-p/d}$. For a 3-dimensional system of N particles, $d = 3N$, and for Simpson's rule $p = 4$ leading to a scaling of $\mathcal{N}^{-1/(0.75N)}$. Then, even for a relatively small many-body system with $N = 10$, doubling the number of function evaluations divides the error by ≈ 1.01 .

4 Diffusion Monte Carlo (DMC)

In the previous section we provided a robust way to produce an upper bound to the ground state energy. Variational Monte Carlo achieves this by minimizing the variational energy (calculated using Monte Carlo techniques) in a sub-space of the system's Hilbert space, which is parametrized by a set of variational parameters α . Depending on the choice of the functional form for the trial wavefunction the minimum of E_V (i.e., the tightest bound), can differ substantially from the ground state energy and approaching further might require substantial insight into the structure of the ground state, which we are looking for in the first place, or a prohibitively large number of variational parameters. In this sense, VMC is not easily systematically improvable.

4.1 Projection via imaginary time-evolution

In this section we will attempt a solution to this problem, that is, identifying the ground state energy, and properties of the ground state, through a clever recasting of Schrödinger's equation. We will describe what will turn out to be Diffusion Monte Carlo, which in principle, is systematically improvable and exact. Diffusion Monte Carlo, relies on the observation that Schrödinger's equation is a diffusion equation in imaginary time. Note that, up to here, we have been concerned with the time-independent Schrödinger's equation and this will remain the case; we're only using the time dependence as a trick to gain insight in the system's ground-state.

We start from the time-dependent Schrödinger's equation for a many-body state $|\Psi\rangle$,

$$i\hbar\partial_t |\Psi(t)\rangle = (H - E_T) |\Psi(t)\rangle , \quad (54)$$

where E_T is an arbitrary energy offset. Equation 54) has the general solution

$$|\Psi(t)\rangle = U(t) |\Psi(0)\rangle \quad (55)$$

where we defined the time evolution operator

$$U(t) = e^{i\frac{t}{\hbar}(H-E_T)} . \quad (56)$$

whose coordinate space representation is a Green's function,

$$G(\mathbf{X} \leftarrow \mathbf{X}', t) = \langle \mathbf{X} | e^{i\frac{t}{\hbar}(H-E_T)} | \mathbf{X}' \rangle \quad (57)$$

which we can integrate with the many-body wavefunction at time t_0 over space to get the wavefunction's value at a time $t' = t_0 + t$:

$$\Psi(\mathbf{X}, t') = \int d\mathbf{X}' G(\mathbf{X} \leftarrow \mathbf{X}', t) \Psi(\mathbf{X}', t_0) . \quad (58)$$

We defined this object because we will soon move from the abstract representation with kets to a coordinate space representation which is closer to practical applications. Making the substitution $t = i\tau$, we get

$$-\hbar\partial_\tau |\Psi\rangle = H |\Psi\rangle , \quad \text{and} \quad U(t) = e^{-\frac{\tau}{\hbar}(H-E_T)} = \bar{U}(\tau) . \quad (59)$$

That is in imaginary time, Schrödinger's equation becomes a diffusion equation. This means that the imaginary time evolution of any initial state $|\Psi_T\rangle$ will eventually converge on the eigenstate of the Hamiltonian with the lowest energy (which is not orthogonal to the initial state). This is easy to show using an expansion like we did in Eq. (31):

$$|\Psi(\tau)\rangle = \bar{U}(\tau) |\Psi_T\rangle = \bar{U}(\tau) \sum_{n=0}^{\infty} a_n |\Phi_n\rangle = \sum_{n=0}^{\infty} a_n e^{-\frac{\tau}{\hbar}(E_n-E_T)} |\Phi_n\rangle \rightarrow a_0 e^{-\frac{\tau}{\hbar}(E_0-E_T)} |\Phi_0\rangle \quad (60)$$

where we assumed that $E_T \approx E_0$ and that $a_0 = \langle \Phi_0 | \Psi_T \rangle \neq 0$. This property is the basis of DMC. However Eq. (60) is far from how this is implemented in practice. To see that let's first start from the simplest case: the free gas.

The free gas case corresponds to neglecting the interacting terms in the Hamiltonian in Eq. (17). This in turn corresponds to the following Schrödinger's equation:

$$-\hbar \partial_\tau |\Psi\rangle = T |\Psi\rangle \quad , \quad \text{or in coordinate space} \quad -\hbar \partial_\tau \Psi(\mathbf{X}, \tau) = \frac{1}{2} \sum_{i=1}^N \frac{\hbar^2}{2m} \nabla_i^2 \Psi(\mathbf{X}, \tau) . \quad (61)$$

If we interpret Φ as a *density* then this is a $3N$ -dimensional diffusion equation. Note that of course Φ is not a density, or a probability density, but rather a probability amplitude, and we need a modulus squared to get a probability density. This is a major issue because it means that unlike any kind of density Φ can become negative when describing fermions and the interpretation of Eq. (61) as a diffusion equation is incomplete. This is the origin of the fermion sign problem. We will ignore it for now, lay the groundwork for DMC and come back to it later where we will see that all of it applies in each pocket of $3N$ -dimensional space where Φ has a definite sign. The Green's function of a diffusion equation is

$$G(\mathbf{X} \leftarrow \mathbf{X}', \tau) = (2\pi\tau)^{-\frac{3N}{2}} \exp \left[-\frac{m|\mathbf{X} - \mathbf{X}'|^2}{2\hbar\tau} \right] . \quad (62)$$

Now we reintroduce the interacting terms in the Hamiltonian, that is we take the entire Hamiltonian from Eq. (17), and the form of the Green's function depends on the form of the two-body, three-body, etc., potentials and in principle does not have a closed form. However, we can write the form of the Green's function for a small time-step τ using the so-called Trotter-Suzuki decomposition which states that

$$\exp[-\tau(A + B)] = \exp[-\tau B/2] \exp[-\tau A] \exp[-\tau B/2] + \mathcal{O}(\tau^3) , \quad (63)$$

which is based on the Zassenhaus formula,

$$\exp[\alpha(A + B)] = \exp[\alpha A] \exp[\alpha B] \exp[-\alpha^2[A, B]/2] \exp[\alpha^3(2[B, [A, B]] + [A, [A, B]])] \cdots \quad (64)$$

Note that in the Trotter-Suzuki decomposition, the term that is second order in τ cancels. Using this decomposition, ignoring higher-order terms, and taking advantage of the simplicity of V 's action on any position state $\mathbf{R}$ we get

$$G(\mathbf{X} \leftarrow \mathbf{X}', \tau) = \langle \mathbf{X} | e^{-\tau(T+V-E_T)} | \mathbf{X}' \rangle = e^{-\tau[V(\mathbf{X})-E_T]/2} \langle \mathbf{X} | e^{-\tau T} | \mathbf{X}' \rangle e^{-\tau[V(\mathbf{X}')-E_T]/2} \quad (65)$$

$$= (2\pi\tau)^{-\frac{3N}{2}} \exp \left[-\frac{m|\mathbf{X} - \mathbf{X}'|^2}{2\hbar\tau} \right] e^{-\tau[V(\mathbf{X})+V(\mathbf{X}')-2E_T]/2} \quad (66)$$

In practice one starts by describing the wavefunction as a collection of M_w walkers at positions $\mathbf{X}_k$, for $k = 1, M_w$, with a density representing the value of the *wavefunction* (*not* the square of the wavefunction), i.e., more walkers where the wavefunctions if larger. The Green's function is then interpreted as a probability density for the evolution of the walkers which then evolve until convergence. Since we only have access to the short-time propagator of Eq. (66), the evolution must be done in very small time-steps τ keeping the error from the Trotter-Suzuki approximation in Eq. (63) (which goes like τ^3 under control. Moreover, the final term in the short-time Green's function in Eq. (66) allows us to interpret the propagation as that of a free system (i.e., with the Green's function of Eq. (62) where the Green's function gets renormalized at every step by a factor $P = \exp[-\tau(V(\mathbf{X})_V(\mathbf{X}') - 2E_T)]$. In the analogy with the diffusion equation, the potential acts as a source/sink of walkers, depending on the sign of the potential (attractive/repulsive). This interpretation allows one to use a birth/death approach where the free Green's function is sampled instead from Eq. (62) and a renormalization is applied by evaluating P and killing a walker with probability $1 - P$. If $1 - P < 0$ a walker is created with probability $P - 1$. Equivalently one can use attach a weight to each walker and accumulate the product.

4.2 Importance sampling

We presented above a simple algorithm for the implementation of DMC. However, in practice this is very inefficient. The efficiency of DMC can be greatly improved by introducing importance sampling which works similarly to the method by the same name in Monte Carlo quadrature: one uses a priori knowledge of the outcome to achieve better sampling.

The underlying idea of importance sampling is that we might have some intuition as to what the system's ground state will look like and we use that as a trial wavefunction. In practice these are low-resolution descriptions of the system, e.g., a mean-field description, or a model that captures the expected structure qualitatively. In general, the trial wavefunction is thought as “almost” the true ground state of the system and we want to find what's missing. Following this logic it makes sense to define a function $f(\mathbf{X}, \tau) = \Phi(\mathbf{x}, \tau)\Psi_T(\mathbf{X})$, where $\Phi(\mathbf{X}, \tau)$ is a wavefunction that satisfies the imaginary time Schrödinger's equation in Eq. (59). The time evolution of $f(\mathbf{X}, \tau)$ then is

$$-\partial_\tau f(\mathbf{X}, \tau) = -\frac{1}{2} \sum_{i=1}^N \nabla^2 f(\mathbf{X}, \tau) + \sum_{i=1}^N \nabla \cdot [\mathbf{v}_D^{(i)} f(\mathbf{X}, \tau)] + [E_L(\mathbf{X}) - E_T] f(\mathbf{X}, \tau) \quad (67)$$

where

$$\mathbf{v}_D^{(i)} = \nabla_i \ln |\Psi_T(\mathbf{X}, \tau)| = \frac{\nabla_i \Psi_T(\mathbf{X})}{\Psi_T(\mathbf{X})}, \quad E_L(\mathbf{X}) = \frac{H\Psi_T(\mathbf{X})}{\Psi_T(\mathbf{X})}. \quad (68)$$

The quantity $\mathbf{v}_D^{(i)}$ is called the *drift velocity* as it represents a velocity terms that carries walkers towards regions where $\Psi_T(\mathbf{X})$ is larger (note that $\mathbf{v}_D^{(i)}$ is proportional to the momentum of particle i in a many-body system described by $\Psi_T(\mathbf{X})$). Since we expect $\Psi(\mathbf{X})$ to be similar to the true ground state this provides better sampling in regions of space that matter. You might also recognize $E_L(\mathbf{X})$ as the local energy introduced for VMC.

The short-time Green's function for Eq. (67) is $\tilde{G}(\mathbf{R} \leftarrow \mathbf{R}', \tau)$ and it is a modified version of $G(\mathbf{R} \leftarrow \mathbf{R}', \tau)$ in Eq. (66):

$$= (2\pi\tau)^{-\frac{3N}{2}} \exp \left[-\frac{m|\mathbf{X} - \mathbf{X}' - 2\tau\mathbf{V}_D(\mathbf{R}')|^2}{2\hbar\tau} \right] e^{-\tau[E_L(\mathbf{X}) + E_L(\mathbf{X}') - 2E_T]/2} \quad (69)$$

where $\mathbf{V}_D = (\mathbf{v}_D^{(1)}, \mathbf{v}_D^{(2)}, \dots, \mathbf{v}_D^{(N)})$. This new version of the Green's function has many advantages. As already mentioned, the drift velocity carries the walkers towards regions of increasing $\Psi(\mathbf{X})$, improving the sampling in these regions. Additionally, the renormalization term has now turned to $\tilde{P} = \exp\{-\tau[E_L(\mathbf{X}) + E_L(\mathbf{X}') - 2E_T]/2\}$ which greatly diminishes the fluctuations in populations, if we have chosen a good $\Psi_T(\mathbf{X})$. This is because in that case $E_L(\mathbf{X})$ is close to the ground state energy and roughly constant in space. Finally, the drift velocity makes the fixed-node approximation much more efficient because it pushes walkers away from the nodal surfaces where $\Psi_T(\mathbf{X})$ vanishes by definition.